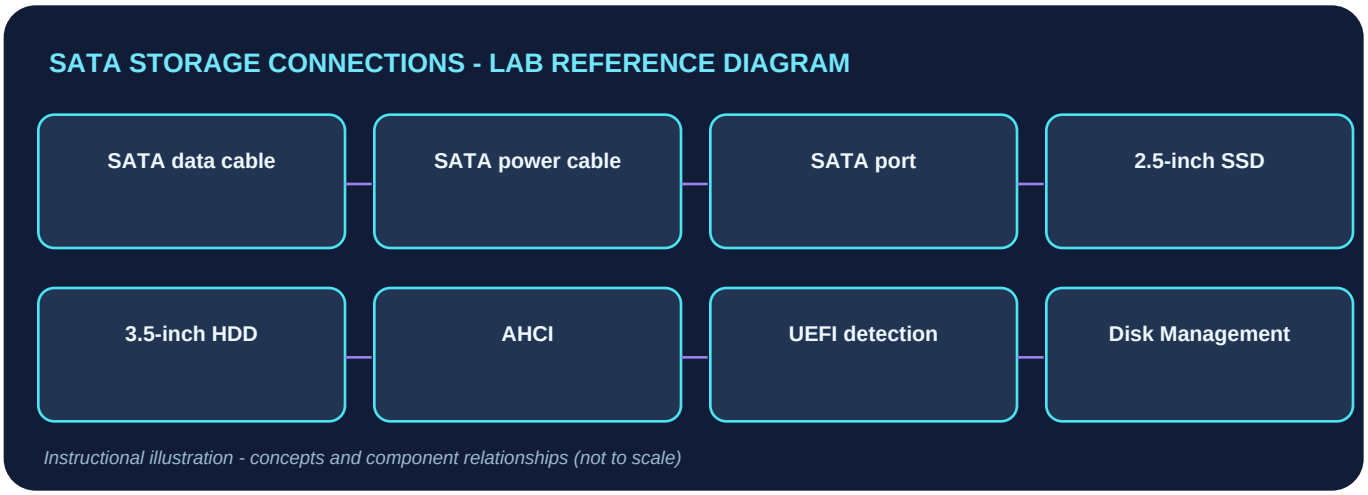
3.2.12 Install SATA Devices

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Connect, detect, and configure SATA storage.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Secure the storage device in a compatible bay.
2. Connect SATA data to the board and SATA power to the PSU.
3. Check drive detection in firmware.
4. Initialize/format only if the lab calls for it.

Troubleshooting and safe practice

• Drive absent: check both cables and controller settings.

• Unallocated drive: use Disk Management without overwriting data.

3.2.12 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
SATA data cable	Cable carrying data between SATA drive and storage controller.
SATA power cable	Power connector supplied from the PSU to a SATA drive.
SATA port	Physical interface or connection point; verify its intended function and compatibility before use.
2.5-inch SSD	A key term in sata storage connections; identify its purpose, where it is used, and how to verify it.
3.5-inch HDD	A key term in sata storage connections; identify its purpose, where it is used, and how to verify it.
AHCI	A key term in sata storage connections; identify its purpose, where it is used, and how to verify it.
UEFI detection	A key term in sata storage connections; identify its purpose, where it is used, and how to verify it.
Disk Management	Windows tool to inspect and manage disks and volumes.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____